

M.E EDUCATION CENTRE

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MATHEMATICS: XII

HALF YEARLY EXAM: 2026

Timings: 2.0 Hr.

Total Marks: 50

Note: (1) Attempt EIGHT (8) Questions from Section: A (3 Marks each) including Q:1 and 2 are compulsory (4 marks each). (2) Attempt FOUR (4) Questions from Section: B (6 Marks each).

SECTION: A

- ✓ 1. Find the derivative by ab-initio: $f(x) = \sqrt{\tan x}$ (Compulsory)
- ✓ 2. A particle moves along the curve $x = 2t^2$ and $y = 4t$. Find the components of velocity and acceleration at $t = 2$ in the direction of $2\hat{i} - \hat{j}$. (Compulsory)
3. Test continuity and discontinuity of the following function: $f(x) = \begin{cases} 2+x, & \text{when } x < 3 \\ 5-2x, & \text{when } x \geq 3 \end{cases}$ at $x = 3$
4. Evaluate the limit of anyone: (i) $\lim_{x \rightarrow e} \frac{\ln x - 1}{x - e}$ (ii) $\lim_{x \rightarrow 1} \frac{\frac{1}{\sqrt{x}} - 1}{1 - x}$
5. Differentiate: $y = \cos^2 x$ w.r.t. $\sin^2 x$.
- ✓ 6. If $y = \sqrt{\tan x} + \sqrt{\tan x} + \sqrt{\tan x} + \dots + \infty$, prove that $(2y - 1) \frac{dy}{dx} = \sec^2 x$. Sec² x.
- ✓ 7. If $y = \tan^{-1} \left(2 \tan^{-1} \frac{x}{2} \right)$, then prove that $\frac{dy}{dx} = 4 \left(\frac{1+y^2}{4+x^2} \right)$ 1 / (2 tan x)
8. Show that $f(x) = \tan^2 x$ is decreasing at $x = \frac{3\pi}{4}$. t ≠ 1, t > 0.
9. For what value of t , the vector function $\vec{r}(t) = \ln(t+3)\hat{i} + \frac{1}{t-1}\hat{j} + \frac{1}{e^t}\hat{k}$ is continuous.
10. Find the area above the x-axis under the following curve, between the given ordinates. $y = 6 \sin^2 x$, $x = 0, x = \frac{\pi}{3}$
11. The x-intercept of a line is the reciprocal of its y-intercept and line pass through $(2, -1)$. Find its equation.
- ✓ 12. The gradient of one of the lines $ax^2 + 2hxy + by^2 = 0$ is twice that of the other. Show that $8h^2 = 9ab$.

$$\frac{-m + \sqrt{h^2 - ab}}{b} \quad \text{or} \quad \frac{-m - \sqrt{h^2 - ab}}{b}$$

SECTION: B

- ✓ 13. (i) Obtain the first three terms of the Maclaurin's series for $f(x) = \cos x$.
 (ii) Find the first four terms of the Taylor's series for the function $f(x) = \sin x$ at $a = \frac{\pi}{4}$.
- ✓ 14. Find the derivatives of any two: (i) $y = \tan^{-1} \left(\frac{\sin 2x}{1 + \cos 2x} \right)$ (ii) $y = \ln \left[\tan \left(\frac{x}{2} + \frac{\pi}{4} \right) \right]$ (iii) $x^y \cdot y^x = 1$
- ✓ 15. Evaluate any two: (i) $\int \frac{1}{x + \sqrt{x}} dx$ (ii) $\int \frac{\sin^3 x}{\sqrt{\cos x}} dx$ (iii) $\int \frac{x^5 dx}{\sqrt{x^2 - 9}}$
16. Evaluate any two: (i) $\int \sec^3 x dx$ (ii) $\int \frac{(5x+6)dx}{(x+3)(x-2)^2}$ (iii) $\int x \cot^{-1} x dx$
17. Compute by using basic properties of any two: (i) $\int_0^{\frac{\pi}{2}} \cos^3 x dx$ (ii) $\int_{\frac{\pi^2}{4}}^{\frac{\pi^2}{36}} \frac{\sin \sqrt{x}}{\sqrt{x}} dx$ (iii) $\int_{\frac{\pi}{2}}^{\frac{\pi}{4}} \tan^2 x dx$
18. If A (2,5), B (3,7) and C (0,8) are the vertices of a triangle then find the equation of (a) median through A (b) right bisector of side AC